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Fine-Grained Vision Foundation Models for Automated Wind Turbine Inspection

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By

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Fine-Grained Vision Foundation Models for Automated Wind Turbine Inspection

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Abstract — Automated visual inspection of wind turbine blades for structural defects such as erosion, cracks, lightning strike damage, and coating delamination presents a viable alternative to manual survey, but existing methods typically involve training convolutional architectures from scratch on scarce domain-specific data. This work explores the potential of a large-scale visual foundational model (AM-RADIO), adapted with a lightweight detection head, to exceed the performance of standard detectors (YOLOv8/9/10/11, RT-DETR) despite having no turbine-related images in its pretraining set. Two experimental conditions were investigated on five independently collected blade-defect datasets: a per-dataset scenario (train and evaluate on four datasets, evaluate on the same dataset) and a cross-dataset scenario (train on four datasets, evaluate on the fifth). AM-RADIO outperformed the strongest of eighteen conventional architectures on four of five datasets by margins of 0.0218–0.0704 mAP@0.50, but underperformed by 0.0907 on the smallest dataset (210 training images), and on merged cross-dataset training reached 0.8306 mAP@0.50 against 0.785 on the baseline. These results show that foundation-model adaptation outperforms conventional detectors in this domain only above an annotation threshold of a few hundred images, and that combining varied datasets is a practical route to surpassing that threshold without additional labelling.

Keywords — *Detection, Foundation models, Transfer Learning, Generalisation, Turbines.*

I. INTRODUCTION

Wind turbine blades operate under continuous mechanical and environmental stress, accumulating erosion, cracking, lightning-strike damage and coating loss which, if undetected, progress to structural failure or unscheduled outage [1]. Inspection currently relies on rope-access technicians or manual review of UAV imagery; both are slow, costly, and limited by inspector fatigue, motivating automated vision-based defect detection.

Published approaches almost all train convolutional or transformer detectors either from scratch or from generic ImageNet initialisation, on a single small domain-specific dataset [2], [3]. This is a reasonable default given how few annotated blade datasets exist, but it leaves an alternative untested: foundation models pretrained on diverse imagery unrelated to wind energy may transfer better than any architecture trained on scarce in-domain data. Blade inspection is a well-suited testbed for this question precisely because every available dataset is small, defect taxonomies differ between sources, and the sources are rarely combined [4].

This work determines whether AM-RADIO [5], an agglomerative foundation model fitted with a lightweight detection head, outperforms conventional detectors

(YOLOv8–v11, RT-DETR) on blade defect detection, both per dataset and on data pooled across sources. Four objectives follow: (i) establish correctly measured conventional baselines across five defect datasets; (ii) fine-tune AM-RADIO per dataset, searching unfrozen backbone depth and backbone-to-head learning-rate ratio; (iii) merge the datasets and test whether foundation-model fine-tuning generalises more effectively than conventional detectors under pooled, heterogeneous data, including on a fully held-out dataset; and (iv) report a methodologically matched comparison in both settings. A secondary objective evaluates whether vision-language models can describe blade defects, zero-shot and after supervised fine-tuning.

Two questions are addressed. Q1: does AM-RADIO fine-tuning outperform conventional detectors when trained and evaluated on each dataset's own splits, and does any advantage hold across datasets of differing size? Q2: does merged cross-dataset training improve AM-RADIO's performance and generalisation by a margin exceeding that achieved by conventional detectors under the same regime?

Both families are trained to convergence under matched early-stopping criteria and evaluated under a common protocol. The scope is five blade defect datasets, one foundation backbone, and a fixed set of conventional baselines; edge-hardware latency and alternative foundation-model families are outside it.

Section II reviews related work; Section III details datasets, architecture, training and tuning; Section IV reports results; Section V discusses findings and limitations; Section VI concludes.

II. RELATED WORK

A. Blade Defect Detection

Shihavuddin et al. [2] created the DTU-Drone UAV dataset and used it to train four Faster R-CNN versions to detect leading edge erosion, lightning receptor damage, and associated surface conditions. Using an Inception-ResNet-V2 architecture, they achieved a mean average precision (mAP) of 81.1%.

While the DTU-Drone dataset has been used extensively since its release, it was initially provided without standardised bounding box annotations. As a result, subsequent researchers have re-annotated the same photographs using their own taxonomic systems; therefore, the results of those subsequent studies cannot be directly compared to each other. To provide a basis for comparison, Gohar et al. [6] created a standardised five-class annotation scheme for the same images.

In addition to providing a consistent annotation scheme for the same set of images, Gohar et al. [7] also introduced slice-aided hyper inference. Slice-aided hyper-inference is a method of processing ultra-high-resolution drone imagery in overlapping window segments instead of fixed-size patches. When applied to small objects, this technique improved small-object mAP@0.50 by up to 13.2 points across YOLOv5, RetinaNet and Faster R-CNN. Yang et al. [8] designed Blade30 to stitch together drone-captured photographs to create full-blade views. Jia and Chen [9] combined optical and thermal video to produce near-real-time segmentation. Both studies suggest that the acquisition and preprocessing components of the overall process are still being refined. Each of these studies trained their convolutional detectors from either scratch or from an ImageNet-trained backbone on a single small corpus. Therefore, each study addressed the resulting data shortage via architecture, i.e., using deeper backbones, sliced inference, instead of altering where the visual prior originated.

Additionally, taxonomic systems developed outside the DTU lineage (e.g., the five-category scheme of the Liu dataset [10]) render different datasets incompatible with each other, thus limiting exploration of cross-dataset training.

B. Conventional Detectors

The baselines used for this evaluation represent the current single stage state-of-the-art. YOLOv8 [11], YOLOv9 [12] and YOLOv10 [13] further enhance the anchor-free, decoupled head architecture of the original YOLO architecture using programmable gradient information and NMS-free end-to-end training. RT-DETR [14] also uses the set prediction capabilities of DETR for real time inference. RT-DETR reports higher COCO accuracy than comparable YOLO variants. Most prior studies on blade inspection have benchmarked their models against older detection architectures. Therefore, the use of these newer detection architectures will provide a more meaningful basis for comparison.

C. Foundation Models

Li et al.'s [15] work shows that a simple, non-hierarchical Vision Transformer (ViT) can serve as a detection backbone, provided a basic feature pyramid is created from a single-scale feature map. This is in contrast to designing a specific hierarchical network for detection.

Therefore, the backbone does not need to be specifically designed for detection; it simply needs to be well pretrained, which is the approach followed here. Ranzinger et al. [5] also focused on pre-training by taking CLIP [16], DINOv2 [17] and SAM [18] and combining them into AM-RADIO. AM-RADIO performs better than all three of its teachers on most benchmarks and also retains their zero-shot and dense-prediction capabilities.

However, AM-RADIO has not been evaluated for defect detection within an unrelated industrial domain. Eltoum et al.'s RadioCore [19] is the closest to evaluating AM-RADIO. They used multi-scale, few-shot memory banks for anomaly segmentation on MVTec-AD and VisA without fine-tuning their RADIOv2.5 backbone. Training-free

feature matching differs from fine-tuning a backbone with a trainable head. Bottles and circuit boards are much more similar to web-image pre-training data than aerial close-ups of composite blade surfaces are. WinCLIP [20] achieved 91.8% AUROC on MVTec-AD by using a compositional prompt ensemble over CLIP embeddings, again on manufactured objects. Section IV-E reports a replication on blade imagery. Currently, there is no research comparing agglomerative foundation models with trainable detection heads for wind turbine blade defect detection to conventional detectors under both single-dataset and merged training.

III. METHODS

A. Datasets

Table I.
DATASET COMPOSITION AND CLASS COVERAGE

Dataset	Train	Val	Test	Classes
Blade30	210	-	93	5/6
Gohar	423	-	93	5/6
Zhang	654	140	139	2/6
WTBD	852	213	-	6
Yae	2,314	-	495	5/6
Merged	4,243	931	Held Out	6

Five independently collected blade defect datasets were used: Blade30 [8], Gohar [6], [7], WTBs2025 [21], WTBD [22] and Yae; derived from Liu and Liu [10]. Each arrived in a different annotation format and was converted through a shared pipeline into YOLO-format bounding boxes under one six-category taxonomy (crack, erosion, lightning receptor damage, paint-off, missing teeth, contamination), so that conversion differences could not confound the comparison. Coverage of that taxonomy varies (Table I). Blade30 lacks missing-teeth instances and Gohar lacks contamination; Gohar's source indices were remapped as 0 = missing teeth, 1 = erosion, 2 = lightning receptor damage, 3 = crack, 4 = paint-off. Label correctness was verified across all five datasets by matching each converted image to its raw source and comparing derived classes against those written by the conversion pipeline; cross-dataset duplication was checked by perceptual hashing. Only one required exclusions; WTBs2025 [21] (referred to as 'Zhang' in this paper) provides 7,544 images across nine defect categories. Three categories had no equivalent in the shared taxonomy and were excluded. Lightning strike was excluded because the released data combines real photographs with GA-DCGAN-synthesised samples that could not be reliably separated. Manual review of the remaining categories found annotation reliability varied considerably: sampled crack and erosion boxes frequently did not contain the labelled defect, and both categories were excluded. Coating detachment was retained only after individual review of all 1,085 images, of which 161 were confirmed as showing the labelled defect. Contamination, verified as reliable in sampling, was retained in full. The resulting subset comprises 933 images across two

categories. WTBD populates all six classes but has no independent test split, so its validation split serves as the held-out evaluation set throughout (Section V). Yae, the largest source, uses its own five-category scheme (corrosion, hide-craze, surface-eye, thunderstrike, dirt), mapped onto the shared taxonomy during conversion. Both training and evaluation images were resized to 1024×1024 . These resizing parameters were applied equally to both model types. Augmentation during training in the foundation model pipeline included random affine transformation, colour jitter, horizontal flip, and random erase. Attempted mosaic augmentation was ultimately rejected due to its interaction with the fixed patch size of the frozen backbones. No augmentation occurred during evaluation for either model type. For the cross-dataset experiment, training splits of Gohar, WTBD, Zhang, and Yae were combined into a unified set of 4243 training and 931 validation images.

Blade30 was excluded from this pool entirely and retained as an unseen source, allowing merged training to be tested on a dataset contributing nothing to it

B. Model Architecture

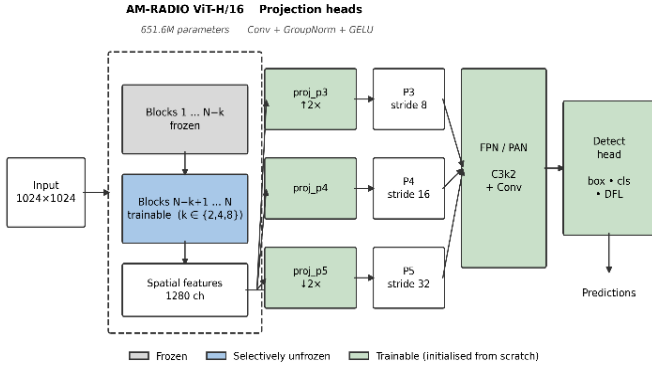


Fig. 1. AM-RADIO detection architecture. The frozen ViT-H/16 backbone supplies spatial features to three trainable projection heads forming a P3/P4/P5 pyramid, followed by an FPN/PAN neck and Detect head. k denotes the number of unfrozen final transformer blocks, tuned per dataset.

Two model families were compared. The conventional baseline comprised eighteen architectures from the YOLO and RT-DETR families (YOLOv8n/s/m/l, YOLOv9c/e/s/m, YOLOv10n/s/m/l, YOLOv11n/s/m/l, RT-DETR-l/x), each loaded from Ultralytics' official COCO-pretrained checkpoints and fine-tuned separately per dataset. Starting both families from large-scale pretrained weights isolates the pretraining corpus and adaptation strategy rather than confounding them with the presence or absence of pretraining.

The foundation-model detector pairs a fixed AM-RADIO ViT-H/16 backbone [5] with a trainable detection head (Fig. 1). Spatial features are extracted at constant patch resolution and mapped into a three-level pyramid at strides 8, 16 and 32 through three lightweight projection blocks, following the plain-backbone approach of Li et al. [15]. Each block applies two convolutional layers with GroupNorm (32 groups) and GELU activation. GroupNorm replaced batch normalisation because backbone memory limits the batch size to four, at which BatchNorm accumulates unstable running statistics; the training-

evaluation discrepancy reached 0.28 to 0.38 mean absolute difference before the change, and normalising independently of batch statistics removed it.

A configurable number of the backbone's final transformer blocks are unfrozen so the pretrained representation can adapt, tuned as described in Section III-D. The three pyramid levels feed a feature pyramid and path aggregation network built from Ultralytics' C3k2 and Conv modules, followed by an Ultralytics Detect head trained with the YOLOv8 detection loss. Sharing the neck and head means the two families differ in backbone and adaptation strategy alone, which is what the comparison isolates.

C. Training Procedure

The baseline models were developed using Ultralytics' default training loop which consists of AdamW-family optimisation with framework-selected configuration, batch sizes of 8, up to 100 epochs with early stopping after 20 consecutive epochs without improvement, cosine learning rate scheduling and framework-default Exponential Moving Average (EMA) weight selection for validation and checkpointing.

The foundation-model pipeline uses AdamW with four separate groups of parameters (two for the unfrozen backbone blocks and two for the newly initialised head). The decayed parameters are separated from the non-decayed parameters. Parameters associated with bias and normalisation layers are excluded from weight decay. In addition, the learning rate for the backbone is reduced relative to that of the head and is discussed further in Section III-D. A base learning rate of 1×10^{-3} and weight decay of 5×10^{-4} are used in conjunction with a three-epoch linear warm-up period followed by a cosine decay of 1% of the original learning rate by epoch 100. Early stopping criteria are identical to those of the baseline model, i.e., evaluation every five epochs until there has been 20 consecutive epochs without improvement. Additionally, Automatic Mixed Precision with Gradient Scaling is employed throughout the training process with gradient norm clipping at 10.0.

An EMA of trainable parameters is employed to aid in model selection and checkpointing. The decay factor is 0.9999 and the time constant is 2000 updates. These are Ultralytics' default values for EMA; therefore, both families employ similar methods to select checkpoints. As such, neither family is afforded an unfair advantage due to differing checkpoint selection rules. EMA is applied only to trainable parameters, as applying EMA to the frozen backbone will produce no benefit.

Although both families use the same early stopping criteria to determine convergence, there exists an asymmetry between them. Ultralytics aggregates gradients to a nominal batch size of 64 while the foundation-model pipeline takes an optimiser step per batch at a batch size of 4, representing a sixteen-fold difference in effective batch size. Section V examines the observed effects of this difference.

D. Hyperparameter Tuning

A grid search was run per dataset over the two hyperparameters governing backbone adaptation: the number of the backbone's final transformer blocks left unfrozen (2, 4 or 8), controlling how much of the pretrained representation can change; and the backbone-to-head learning rate ratio (20, 50 or 100), which follows standard discriminative fine-tuning practice in preventing early large gradients from erasing useful pretrained features. This gives nine configurations per dataset, each trained under identical early-stopping criteria, with the highest EMA-selected validation mAP@0.50 retained.

The remaining hyperparameters (input resolution, augmentation policy, scheduling length and clipping threshold) remained constant across all runs of the grid search, so that any differences observed in performance can be solely attributed to the two adaptation hyperparameters. Varying them jointly would have required a substantially larger search than the available compute allowed.

Three of the five datasets, Gohar, Zhang and Yae, performed best with four unfrozen blocks. Yae (the largest) continued to show that four unfrozen blocks were preferred, demonstrating that adaptation depth does not increase proportionally with the total number of images in the training set. This is revisited later in Section V as it appears that both annotation density and class coverage matter alongside the sheer volume of images.

To execute the cross-dataset experiment, the best performing configuration from each individual dataset's grid search was used as opposed to running another grid search because of the cost of running a single configuration at the scale of the combined dataset. All nine configurations for each dataset's grid search were trained using identical batch sizes, accumulation schedules, scheduling lengths and clipping thresholds; therefore, comparisons among configurations are internally controlled and not affected by the pipeline asymmetry described above.

E. Vision-Language Model Investigation

A second stream of research examined whether vision-language models could classify blade defects based upon cropped portions of images; these were combined with the localising capabilities of the detector by providing a textual explanation for the identified defect. Five different prompt methods were used to test the ability of several general purpose vision-language models (BLIP-2 in both OPT and FlanT5-XL configurations, Qwen2.5-VL-3B-Instruct, and a reproduction of the WinCLIP anomaly classifier based on CLIP [20]) to detect blade defects. The results of this initial testing are presented in Section IV-E, and they provided the rationale for performing the supervised fine-tuning discussed next.

Fine-tuning used Qwen3-VL-2B-Instruct and applied LoRA (low-rank adaptation, rank=16, alpha=16) to the vision, language, attention and MLP layers inside the Unsloth framework. This updated about 1.1% of the total number of parameters in the model (2.15 billion). The training dataset consisted of 6,924 labelled crops extracted from all five blade datasets. All crops had 50% padding added around the annotated bounding box, and had a

minimum crop dimension of 100 pixels, allowing the surrounding blade surface to be present as contextual information. Additionally, 1,500 negative crops were generated by sampling from source image corners, giving the model non-defective regions to differentiate from. Finally, each crop was transformed into a single turn image and instruction pair where the ground truth class name was used as the desired output response.

Training was performed for three epochs using 8-bit AdamW at a learning rate of 2×10^{-4} with linear decay and five warmup steps, an effective batch size of 8, and mixed precision. At the end of training, a final training loss of 0.018 was achieved. Evaluation of the trained adapters was conducted using both the Gohar test set and WTBD validation split, which were kept separate from the fine-tuning process.

F. Ethical Considerations

All datasets used in this study are secondary, publicly available or previously published research datasets consisting of aerial and ground-based photographs of wind turbine infrastructure; none contain personally identifiable information; no primary data collection involving human participants was undertaken; thus no consent procedures were necessary. Where datasets are derived from a shared underlying image source, original dataset and annotation provenance is credited in Section III-A and in the corresponding dataset descriptions above. As the foundation model backbone used was pretrained on broad general-purpose web imagery unrelated to wind energy; its underlying training data provenance and any associated biases are inherited rather than independently assessed within this study; this is identified as a limitation in Section V. Given the safety-critical nature of wind turbine inspection, validation well beyond the scope of this study would be required before any of these approaches could be used in place of, rather than alongside, qualified human inspection. Fairness across defect categories is a further concern: pronounced class imbalance means per-class reliability is unequal, and the weakest classes in this study include structurally critical damage, so deployment would require per-class validation rather than aggregate accuracy.

IV. EXPERIMENTATION AND RESULTS

A. Hardware and Software Environment

All experiments were conducted on Google Colab using a single NVIDIA A100-SXM4 GPU with 40 GB of memory. The software environment comprised Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, Ultralytics 8.4.145, and torchmetrics for detection metric computation. AM-RADIO ViT-H/16 checkpoints were obtained through torch.hub. The vision-language experiments used the HuggingFace transformers library with Salesforce's BLIP-2 checkpoints, Qwen2.5-VL-3B-Instruct, OpenCLIP with LAION-400M pretraining, and Qwen3-VL-2B-Instruct fine-tuned through the Unsloth framework with low-rank adaptation. A full environment specification is provided in the supporting material.

Conventional baselines were trained through Ultralytics' native pipeline with automatic mixed precision and

framework-default augmentation. The foundation-model pipeline used a custom training loop implementing the procedure described in Section III-C, sharing the Ultralytics detection head, loss function and non-maximum suppression implementation so that the two families differ only where intended

B. Evaluation Metrics

Detection performance is reported as mean average precision at an IoU threshold of 0.50 (mAP@0.50), alongside mAP averaged over 0.50 to 0.95 in 0.05 increments. The former is the standard detection metric and Ultralytics' internal selection criterion, allowing consistent comparison between families; the latter penalises imprecise localisation, where a plain ViT backbone with a lightweight projection pyramid might reasonably differ from a purpose-designed convolutional detector. Vision-language results are reported as classification accuracy alongside false-positive rate on undefected regions, since accuracy alone cannot distinguish genuine discrimination from a fixed bias toward one class.

C. Metric Limitations

Reporting both IoU thresholds reduces but does not remove the limitations of aggregate metrics. Neither value is a claim of deployment sufficiency, and neither reflects per-class reliability, which Table IV shows varies considerably. The WinCLIP replication in Section IV-E uses AUROC, matching the original publication [20], since a fixed threshold would misrepresent a method never optimised for this task. Several evaluation sets are small: Blade30's test split contains 93 images and some per-class samples number fewer than 15, so reported values carry meaningful sampling variance and small per-class differences should not be over-interpreted.

D. Per-Dataset and Cross-Dataset Detection Results

TABLE II.

Per-Dataset Detection Results (mAP@0.50), Ordered by Training Set Size

Dataset	Train Images	Baseline (model)	AM-RADIO	Δ
Blade30	210	0.4226 (YOLOv8s)	0.3319 (blocks 2, ratio 20)	-0.0907
Gohar	423	0.8335 (RT-DETR-x)	0.8553 (blocks 4, ratio 100)	+0.0218
Zhang	654	0.4258 (YOLOv9e)	0.4524 (blocks 4, ratio 50)	+0.0266
WTBD	852	0.8371 (YOLOv9s)	0.9075 (blocks 8, ratio 20)	+0.0704
Yae	2314	0.8529 (YOLOv9s)	0.8977 (blocks 4, ratio 50)	+0.0448

Table II reports the best conventional baseline and the best AM-RADIO configuration for each dataset, selected as described in Section III-D.

AM-RADIO exceeded the strongest conventional baseline on four of five datasets, by 0.0218 to 0.0704 mAP@0.50. The exception is Blade30, which

underperformed by 0.0907 and is the only dataset with fewer than 400 training images. Above roughly 400 images the foundation model is competitive or superior; below it the advantage reverses. The margin does not scale linearly with size: WTBD, at 852 images, gained the most (+0.0704), while Yae, with nearly three times as many, gained +0.0448. Dataset difficulty and annotation characteristics evidently matter alongside image count. Zhang is instructive: both families score poorly in absolute terms (0.4258 and 0.4524), yet the relative margin remains positive, so the advantage does not depend on the task being easy.

The dataset on which the foundation model performed worse than the conventional baseline was also the one that required the most conservative modification of the pretrained representation. This is consistent with insufficient data to allow for effective backbone fine-tuning.

The cross-dataset study represents the most direct evaluation of whether combining heterogeneous datasets can address the limitation described above.

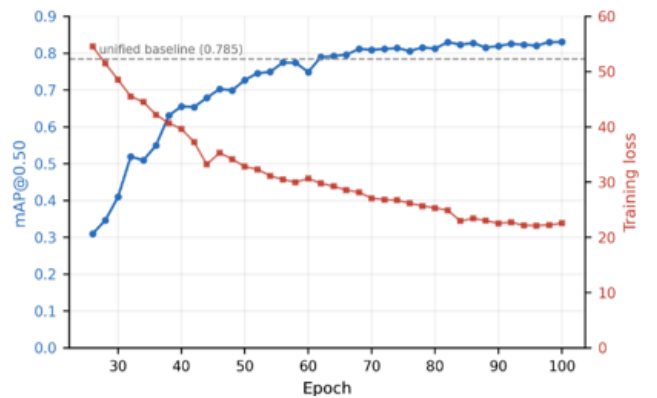


Fig. 2. Training progression on the merged cross-dataset experiment. mAP@0.50 crosses the baseline at epoch 62 and continues to improve as training loss falls, reaching a final mAP@0.50 of 0.8306

TABLE III.

CROSS-DATASET GENERALISATION (mAP@0.50)

Test Set	Baseline (Per-dataset)	Baseline (Merged)	AM-RADIO (Per-dataset)	AM-RADIO (Merged)
Gohar	0.8335	0.7443	0.8553	0.8200
Zhang	0.4258	0.4153	0.4524	0.4302
WTBD	0.8371	0.8611	0.9075	0.8863
Yae	0.8529	0.8272	0.8977	0.8725

Table III decomposes both merged models across the individual test sets. On the merged validation split AM-RADIO reached 0.8306 mAP@0.50 against 0.785 for a YOLOv9s baseline trained on the same data. Table III shows the advantage holds on every individual test set, so it is not an artefact of how the pooled split is composed. Both families score below their per-dataset counterparts, as expected when one model must fit four independently collected sources, though the foundation model gives up

roughly half as much (0.0260 mean reduction against 0.0459).

E. VLM Results

Fine-tuning Qwen3-VL-2B via low-rank adaptation on 6,924 domain-specific crop images yielded 57.3% classification accuracy on the Gohar test split and 86.5% on the WTBD validation split; this contrasts sharply with approximately 14% achieved when randomly selecting one of the seven possible answers. Neither split contributed to fine-tuning. Differences in accuracy between these two splits are primarily due to their different class compositions, specifically, WTBD has a relatively balanced distribution of all six categories of defects, whereas Gohar's crop images are heavily weighted towards two categories and have very few examples in some other categories.

False positive rates for the negative crop images (sampled from areas where no defects were identified) were 11.8% on Gohar and 16.0% on WTBD, showing that the model correctly differentiates defective blade surfaces from non-defective ones in most instances. This cannot be determined based solely upon accuracy since a model that consistently predicts a defect would appear successful if evaluated solely on a dataset containing only defects.

Table IV.

PER-CLASS CLASSIFICATION ACCURACY OF THE FINE-TUNED VISION-LANGUAGE MODEL ON HELD-OUT DATA

Class	Gohar (test)	WTBD (val)
Crack	72.7% (8/11)	93.1% (149/160)
Paint-off	70.0% (7/10)	87.1% (74/85)
Erosion	58.6% (34/58)	69.8% (37/53)
Lightning receptor damage	50.0% (3/6)	71.4% (10/14)
Missing teeth	48.7% (19/39)	absent from split

Per-class results in Table IV show uneven reliability, from 93.1% on cracks in WTBD to 48.7% on missing teeth in Gohar, with the weakest categories being those with fewest training crops. Missing teeth is among the most structurally consequential defect types, which bears on the fairness considerations in Section III-F. Under-detection rather than misclassification dominates the error profile.

Integrating the fine-tuned model with the detector confirms the two components function together. Using the detector's own predicted boxes rather than ground-truth annotations, detector and VLM agreed on class assignment in every case at detection confidence 0.22 or above. Below that threshold, disagreements clustered by source image rather than by confidence value, indicating the VLM responds to image content rather than defaulting to a fixed answer under uncertainty.

Regarding zero-shot evaluations, it was determined that none of the general purpose vision-language models evaluated without additional training could be used for this domain. Both BLIP-2 configurations produced near-identical outputs for annotated defective crops and matched

control crops from the same source images, across all five prompting strategies.

Although Qwen2.5-VL-3B exhibited a response pattern associated with class-based responses when using structured prompting techniques, altering the order of the same answer options for the same crop resulted in a change in the majority prediction from erosion (18/24) to no observable defect (19/25), indicating that the position of the answer within the list is more influential than the image content itself.

Finally, WinCLIP was replicated using the backbones and ensemble of prompts outlined in the original article and generated an AUROC score of 0.608 relative to the 0.918 reported in the original article's manufactured-object dataset and 0.500 relative to random chance. Fewer than 7,000 domain-specific crops significantly reduced the discrepancy resulting from generalising pre-training alone.

V. DISCUSSION

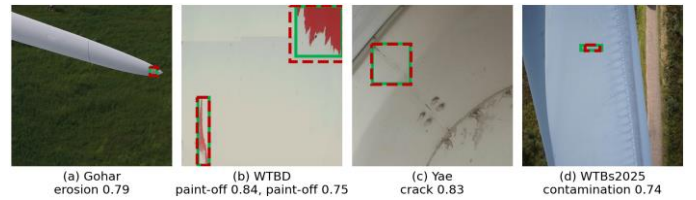


Fig. 3. Detections from the merged cross-dataset model on each contributing source: (a) Gohar, (b) WTBD, (c) Yae, (d) WTBS2025. Ground truth shown as solid green boxes, model predictions as dashed red boxes.

The per-dataset results answer Q1 with a qualification. AM-RADIO outperformed the best of eighteen conventional architectures on four of five datasets, so a fine-tuned foundation model can exceed conventional detectors on this task. The advantage does not hold uniformly, and where it fails is informative: the four successful datasets provided 423 to 2,314 training images, the unsuccessful one 210. Adapting a 651-million-parameter backbone with a newly initialised detection head appears to require a minimum volume of annotated in-domain data, below which a smaller architecture fine-tuned from COCO pretraining is the better choice.

The hyperparameter grid corroborates this independently. Adaptation was most conservative on Blade30, at two unfrozen blocks, so the dataset requiring least backbone modification is also the one where the approach failed. That two independent lines of evidence point to the same threshold strengthens the reading, since the grid was not constructed to test it.

The threshold is not a simple function of image count: adaptation depth and margin size both track annotation density and class coverage as much as they track dataset size.

Q2 is answered affirmatively: the foundation model led on the pooled split and on every source individually. Both families gave up accuracy relative to single-dataset training, but the foundation model gave up around half as much, so

broad pretraining appears to confer tolerance of source heterogeneity alongside the per-dataset gains (Fig 3). This

is the more striking for concerning the same model that proved the weaker choice on the smallest dataset. Whatever makes a large pretrained backbone unsuitable for 210 images is not what governs its behaviour when sources are combined.

The vision-language investigation also demonstrates a similar trend. In all cases for which the use of zero-shot transfer was evaluated, each of the models and prompting techniques were unsuccessful in transferring to the aerial industrial image dataset; the most successful application was a reproduction of WinCLIP, which had an AUROC score of 0.608, slightly better than random and 0.31 lower than when applied to the manufactured objects dataset. With less than 7,000 fine-tuned crop images, almost all of the difference between the two datasets was closed. As such, both parts of this research demonstrate that while general-purpose large-scale pre-training is insufficient to transfer directly to aerial industrial images without additional training data, a relatively small amount of domain-specific training data can enable the model to achieve comparable results.

A. Limitations

Some limitations restrict the generalisability of these findings. As Ultralytics accumulates gradients to an effective batch of 64 and the foundation model pipeline updates gradient per batch of 4, the effective batch sizes for each of the two pipelines differ by a factor of 16. On nearly all iterations, gradient norm clipping at 10.0 occurred, with pre-clip norms varying from approximately 100 to 500. Clipping effectively limits per-step updates regardless of batch size; thus, the effects of the differences in batch size are greatly reduced. Both families of models were trained until convergence using the same early stopping criterion. Therefore, although the use of early stopping does not completely eliminate the possibility of bias due to training time, it substantially reduces it.

Additionally, WTBD does not have a separate test set, therefore the validation set provided by WTBD was used for evaluation purposes for all comparisons. As such, the WTBD results are less rigorously held out from testing than those obtained from the other four datasets.

The merged baseline was YOLOv9s alone rather than the best of the eighteen-architecture sweep used per dataset. Results are also reported from single training runs; checkpoint-to-checkpoint variation reached 0.15 mAP on the smaller datasets, so the Blade30 margin should be read as indicative rather than established.

VI. CONCLUSION AND FUTURE WORK

This work evaluated whether a large-scale visual foundation model, fine-tuned with a lightweight detection head, outperforms conventional detectors on wind turbine blade defect detection. AM-RADIO was compared against eighteen conventional architectures across five independently collected datasets, first per dataset and then under merged training with Blade30 held out entirely. Both families started from large-scale pretrained weights and were trained to convergence under identical criteria.

AM-RADIO exceeded the strongest conventional baseline on four of five datasets, by 0.0218 to 0.0704 mAP@0.50, and underperformed by 0.0907 on the smallest. The failure case is the substantive finding: foundation-model adaptation in this domain appears to require a minimum volume of annotated in-domain data, on the order of several hundred images, below which a smaller architecture fine-tuned from general pretraining is the better choice. The hyperparameter search supports this independently, since the dataset requiring the most conservative backbone adaptation is the one where the approach failed.

A parallel investigation found that general-purpose vision-language models do not transfer to this domain zero-shot, the best replication reaching 0.608 AUROC against 0.500 for chance, but that low-rank fine-tuning of Qwen3-VL-2B on fewer than 7,000 domain crops reached 86.5% classification accuracy on held-out data.

Several questions remain open. It is unclear if the threshold identified is unique to AM-RADIO and its adaptation methodology, or if it represents a common characteristic of all foundation model fine-tuning for detection-type tasks. Only one backbone was evaluated. Similarly, it is unclear why the adaptation depth did not increase linearly as the size of the dataset increased. Annotation density or class composition may affect how optimal adaptation strategies are determined in ways that were not studied in this research.

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REFERENCES

- [1] J. M. Pinar Pérez, F. P. García Márquez, A. Tobias, and M. Papaalias, "Wind turbine reliability analysis," *Renew. Sustain. Energy Rev.*, vol. 23, pp. 463–472, 2013.
- [2] A. Shihavuddin et al., "Wind turbine surface damage detection by deep learning aided drone inspection analysis," *Energies*, vol. 12, no. 4, p. 676, 2019.
- [3] A. Foster, O. Best, M. Gianni, A. Khan, K. Collins, and S. Sharma, "Drone footage wind turbine surface damage detection," in *Proc. IEEE 14th Image, Video, and Multidimensional Signal Processing Workshop (IVMSP)*, Nafplio, Greece, 2022, pp. 1–5.
- [4] I. Gohar, W. K. Yew, A. Halimi, and J. See, "Review of state-of-the-art surface defect detection on wind turbine blades through aerial imagery: Challenges and recommendations," *Eng. Appl. Artif. Intell.*, vol. 144, p. 109970, 2025.
- [5] M. Ranzinger, G. Heinrich, J. Kautz, and P. Molchanov, "AM-RADIO: Agglomerative vision foundation model reduce all domains into one," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 12490–12500.
- [6] I. Gohar, J. See, A. Halimi, and W. K. Yew, "Automatic defect detection in wind turbine blade images: Model benchmarks and re-annotations," in *Proc. IEEE ICMEW*, 2023, pp. 290–295.
- [7] I. Gohar, A. Halimi, J. See, W. K. Yew, and C. Yang, "Slice-aided defect detection in ultra high-resolution wind turbine blade images," *Machines*, vol. 11, no. 10, p. 953, 2023.
- [8] C. Yang, X. Liu, H. Zhou, Y. Ke, and J. See, "Towards accurate image stitching for drone-based wind turbine blade inspection," *Renew. Energy*, vol. 203, pp. 267–279, 2023.
- [9] X. Jia and X. Chen, "AI-based optical-thermal video data fusion for near real-time blade segmentation in normal wind turbine operation," *Eng. Appl. Artif. Intell.*, vol. 127, p. 107325, 2024.
- [10] M. Liu and D. Liu, "Wind turbine blades defect detection based on global and local attention with multi-feature fusion," *Appl. Soft Comput.*, vol. 185, p. 113914, 2025.
- [11] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLO," 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>. Accessed: 10 Sep. 2026.
- [12] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. ECCV*, 2024, pp. 1–21.
- [13] A. Wang, H. Chen, L. Liu, et al., "YOLOv10: Real-time end-to-end object detection," in *Proc. NeurIPS*, 2024.
- [14] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF CVPR*, Seattle, WA, USA, 2024.
- [15] Y. Li, H. Mao, R. Girshick, and K. He, "Exploring plain vision transformer backbones for object detection," in *Proc. ECCV*, 2022, pp. 280–296.
- [16] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.
- [17] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.
- [18] A. Kirillov et al., "Segment Anything," in *Proc. IEEE/CVF ICCV*, 2023, pp. 4015–4026.
- [19] M. A. M. Eltoum, E. Iqbal, S. U. Khan, and Y. Abdulrahman, "RadioCore: Few-shot industrial anomaly segmentation with multi-scale Radio ViT features," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, VISION Workshop, 2026, pp. 7288–7297.
- [20] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, and O. Dabeer, "WinCLIP: Zero-/few-shot anomaly classification and segmentation," in *Proc. IEEE/CVF CVPR*, 2023, pp. 19606–19616.
- [21] R. Zhang, S. Wang, S. Liu, Y. Liang, T. Wang, C. Liu, and S. Chen, "Wind turbine blades surface defect high-quality image dataset construction and performance validation," *Sci. Data*, 2026, doi: 10.1038/s41597-026-07455-1.
- [22] L. Ji, J. Cheng, and S. Wu, "Multiclass dataset for intelligent detection of wind turbine blade defects using drone imagery," *Sci. Data*, vol. 13, no. 1, p. 396, 2026.
- [23] [23] Anthropic, "Claude (Opus 5)," Large language model, 2026. [Online]. Available: <https://claude.ai>

Note: Anthropic's Claude Opus 5 [23] was used to assist with idea generation, debugging and refactoring of the manually authored experimental code for efficiency and style compliance, generate synonyms, and to check the paper for grammatical errors and internal inconsistencies. All experimental work, results, dataset curation, and analysis are the author's own.